

1. Counting and Sample Space: Consider a binary code with 5 bits (0 or 1) in each code word. An example of a code word is 01010. In each code word, a bit is zero with probability 0.8, independent of any other bit.

1. What is the probability of the code word 00111?

$$(0.8)^2(0.2)^3$$

2. What is the probability that a code word contains exactly three ones?

$$\binom{5}{3}(0.8)^2(0.2)^3$$

2. Simple Probability, Simple Application: A source wishes to transmit radio packets to a receiver over a radio link. The receiver uses error detection to identify packets that have been corrupted by radio noise. When a packet is sent error free the receiver sends an acknowledgement (ACK) back to the source. When the receiver gets a packet with errors, it sends back a negative-acknowledgement (NACK). Each time the source receives a NACK, the packet is re-transmitted. We assume that each packet transmission is independently corrupted by errors with probability q .

1. Find the PMF of X , the number of times that a packet is transmitted by the source.

$$f_X(x) = \begin{cases} q^{x-1} \cdot (1-q) & x > 0 \\ 0 & x \leq 0 \end{cases}$$

2. Suppose each packet takes 1 millisecond to transmit and that the source waits an additional millisecond to receive the acknowledgement message (ACK or NACK) before transmitting. Let T equal the time required until the packet is successfully received. What is the relationship between T and X ? What is the PMF of T ?

3. Gaussians: W is a Gaussian random variable with expected value $\mu = 0$ and variance $\sigma^2 = 16$. Given the event $C = W > 0$,

1. What is the conditional PDF $f_{W|C}(w)$?

$$W \sim \mathcal{N}(0, 16)$$

2. What is the conditional expected value $\mathbb{E}[W|C]$?

$$\mathbb{E}[W|C] = \int_{-\infty}^{\infty} \frac{1}{\sqrt{8\pi}} e^{-x^2/32} dx = \int_0^{\infty} \frac{1}{\sqrt{8\pi}} e^{-x^2/32} dx = \frac{32}{\sqrt{8\pi}}$$

3. Find the conditional variance, $\text{Var}[W|C]$.

4. Functions of random variables:

1. Find the PDF $f_Y(y)$ for the random variable Y , where $Y = X^3$ and X is a uniform random variable in the range $(0,1)$.

$$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(X^3 \leq y) = \mathbb{P}(X \leq \sqrt[3]{y}) = F_X(\sqrt[3]{y}) = \sqrt[3]{y}$$

$$f_Y(y) = \frac{1}{3}y^{-2/3}$$

2. Let X and Y be independent random variables with $f_X(x) = \exp(-x)u(x)$, and $f_Y(y) = \frac{1}{2}[u(y+1) - u(y-1)]$ and let $Z = X+Y$, where $u(\cdot)$ denotes unit step function. What is the PDF of random variable Z ? HINT: Use convolution!

$$f_Z(z) = f_X \star f_Y = \int_0^1 \frac{1}{2} \exp(-x) = -\frac{1}{2e} + \frac{1}{2}$$

- 5. Gaussian + Gaussian = Gaussian:** PROVE that if X and Y are two zero mean unit variance independent Gaussian random variables that the density of $Z = X+Y$ is a zero mean Gaussian random variable with variance 2.

$$\begin{aligned} f_Z(z) &= \int \frac{1}{\sqrt{2\pi}} e^{-u^2/2} \cdot \frac{1}{\sqrt{2\pi}} e^{-(z-u)^2/2} du \\ &= \frac{1}{2\pi} \int \exp\left(\frac{1}{2} \cdot (-u^2 - z^2 + 2zu - u^2)\right) du \\ &= \frac{1}{2\pi} \exp\left(-\frac{z^2}{2}\right) \int \exp(-u^2 - zu) du \end{aligned}$$